Spring 2025

ME 571: Medical Robotics

Endoscopic Stabilization Sleeve

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Significance

Context:

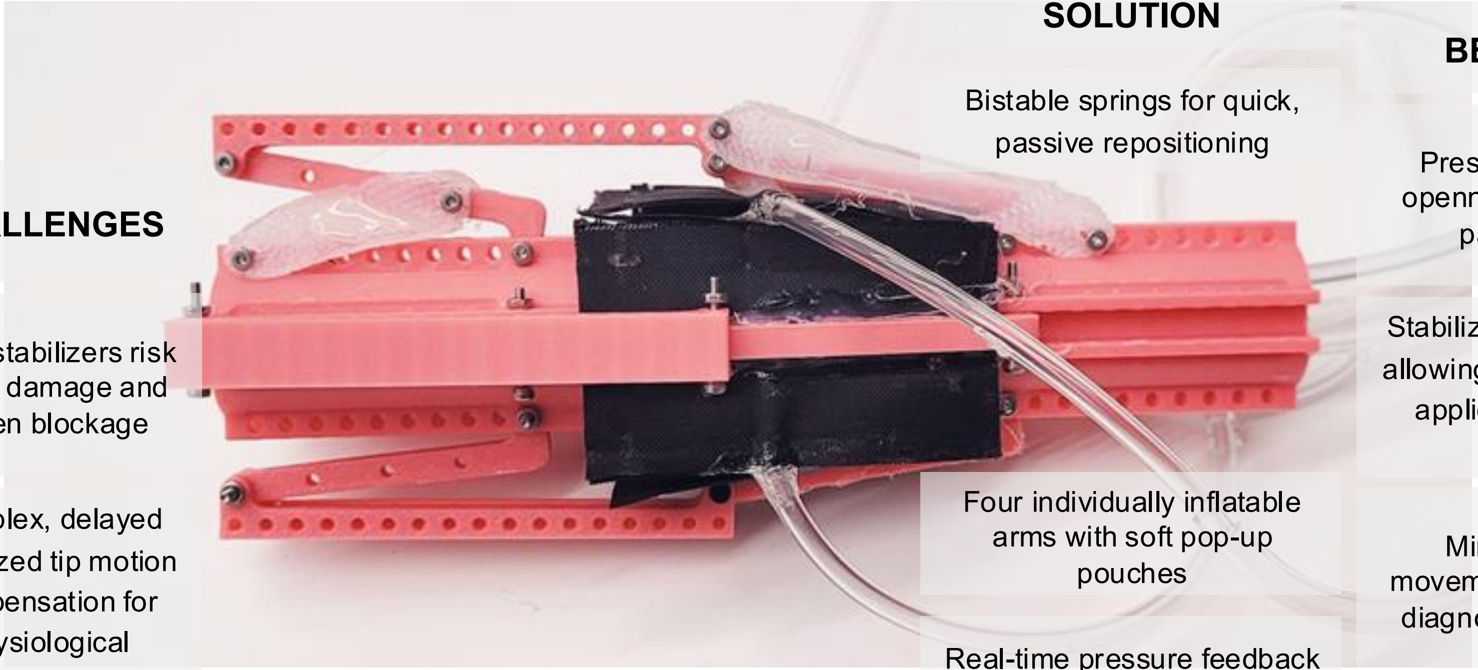
- Cancer is the second leading cause of death in the U.S. [1]
- Early detection increases treatment efficacy and survivability of patients [2]
- Endoscopic tools reduce patient burden and increase diagnostic yield [3]

Challenges:

- Flexible shape may lead to undesired tool pose and force exertion
- Reduce aiming capabilities - increase discomfort and risk to patients
- Difficulty in operation [4]
- Physiological motions of organs create disturbances and motion artifacts [5]

[31]

Innovation



CHALLENGES

Rigid stabilizers risk tissue damage and lumen blockage

Complex, delayed motorized tip motion compensation for physiological movements

SOLUTION

Bistable springs for quick, passive repositioning

Four individually inflatable arms with soft pop-up pouches

Real-time pressure feedback for selective anchoring

BENEFITS

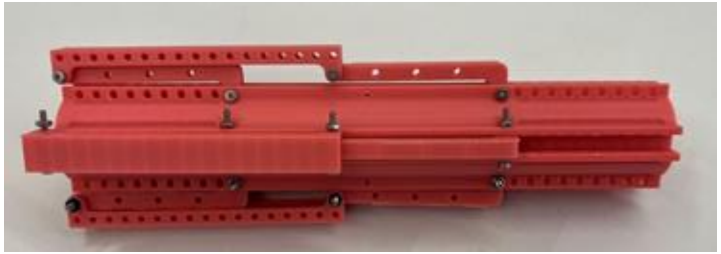
Preserves lumen openness, reducing patient risk

Stabilizes endoscope, allowing stronger force application during biopsy

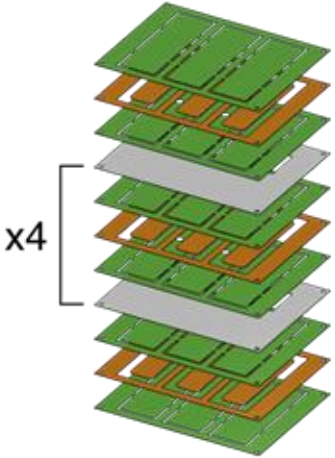
Minimizes tip movement, improving diagnostic accuracy

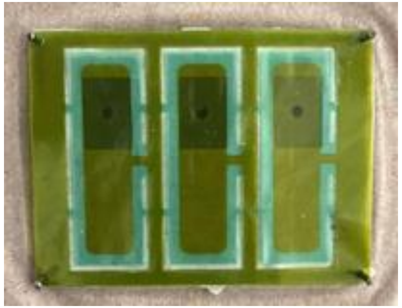
Approach - Endoscope Sleeve and TPE Bellow Production

- Biaxial legs configured in a four bar linkage provides translation [17]



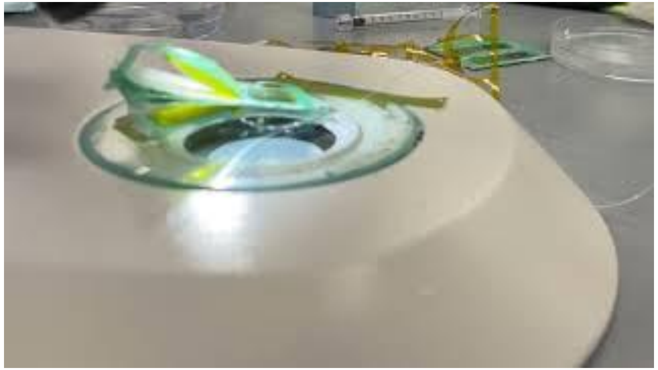
- Bellows originally designed using TPE (Thermoplastic Elastomers), in a layered “pop-up book” MEMS method with kapton and parchment paper separating layers [21]-[23]





Approach - Challenges with TPE Bellows

- Leaking and delamination between bellow layers
- Issues with internal cavity bonding onto itself
- Time consuming manufacturing process



Approach - Heat Weldable Fabric Pouches

- Heat weldable fabric pouches offered more reliability, simplicity, and cut down time
- Pouches resembled triangular prisms when inflated, similar to the TPE bellow design

SUCCESS!

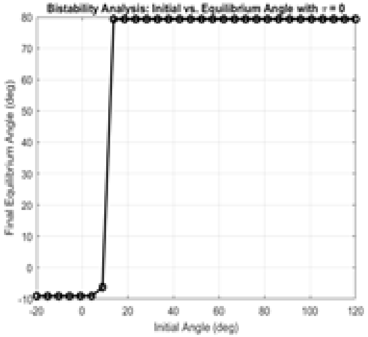
Approach - Modeling Dynamics

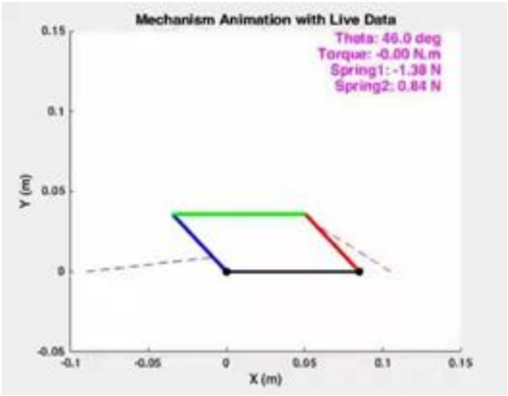
4 bar linkage dynamics simulated in MATLAB

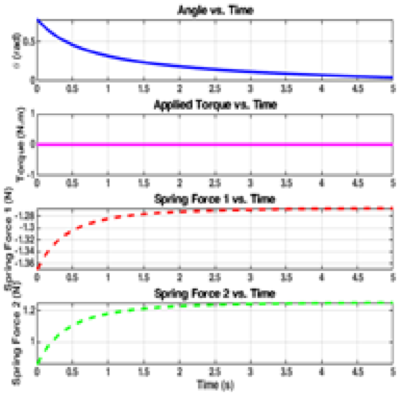
Tested different spring sizes and placements using approximated ecoflex 30 young's modulus

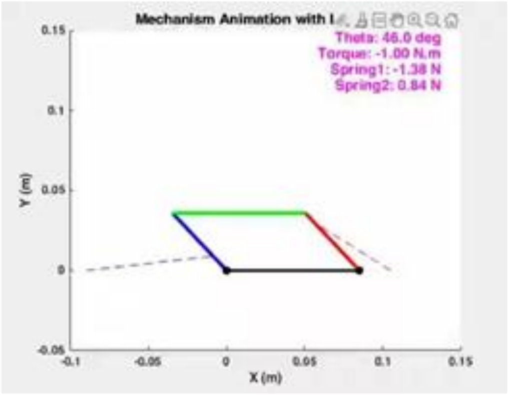
Determined optimal Spring fabrication dimensions + placement for bistability

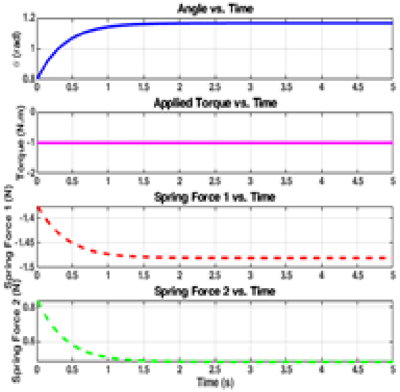
0° (smaller spring) and 90°(larger spring configuration selected









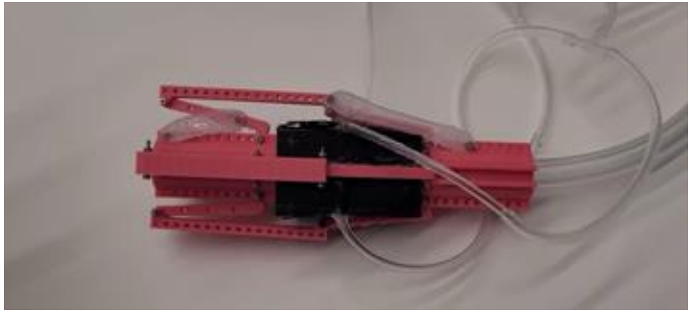


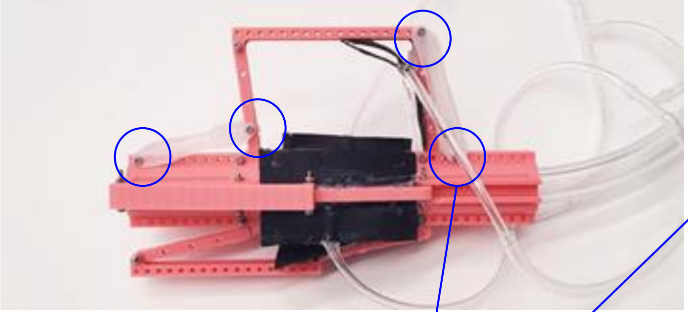
Approach - Spring Development

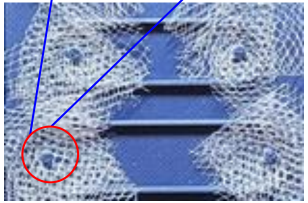
Fabricated springs based on matlab calculations: Silicone casting in 3D printed molds

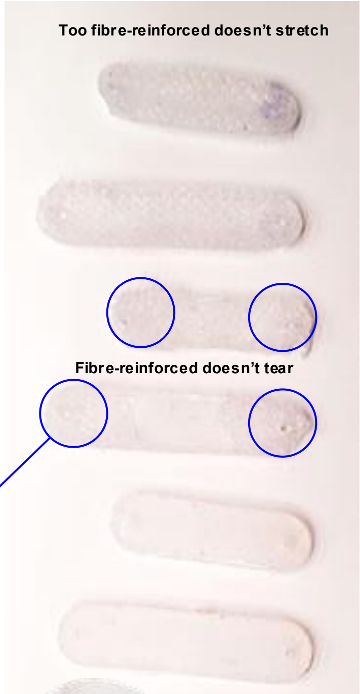
Oversized spring folding mitigated by repositioning smaller spring farther from pivot

Material tearing at attachment points: DS10 + Ecoflex 30 didn't work, solved by fiber-reinforced silicone casting









Non fibre-reinforced easily tears

Approach - Controls

Arduino Uno

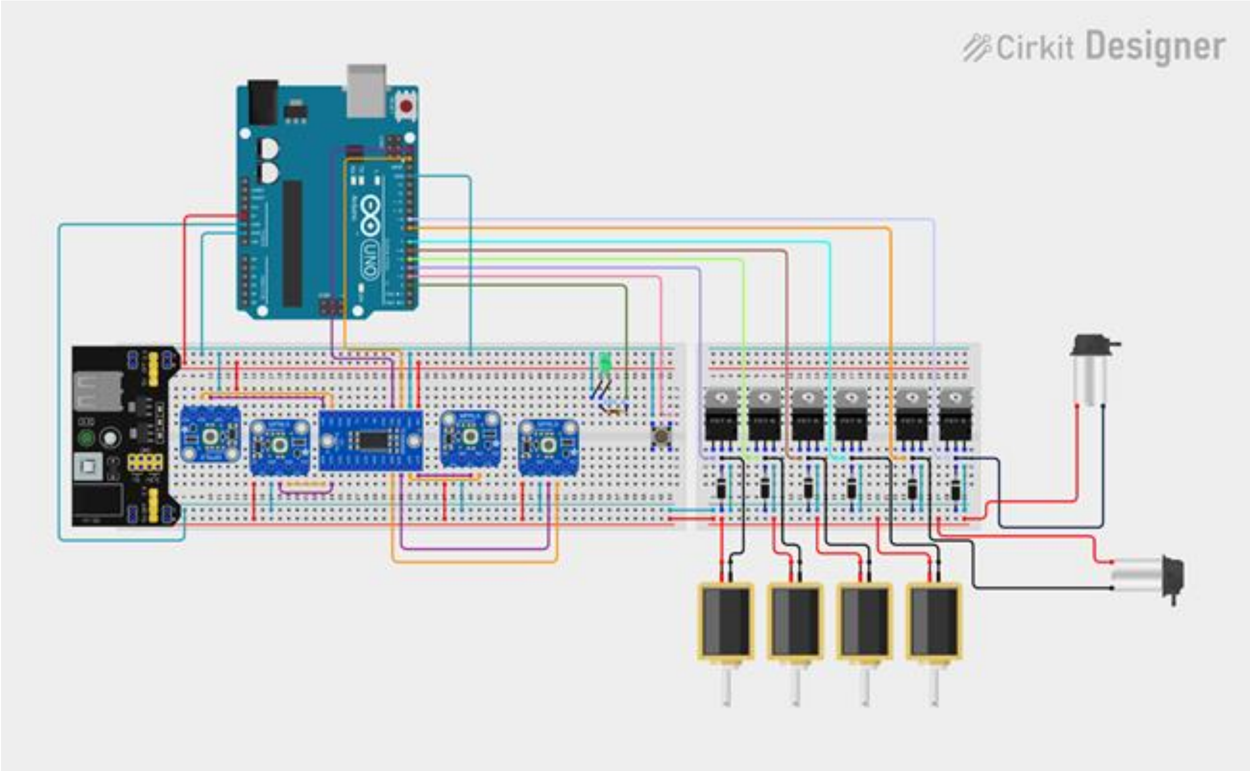
Power module

Mini Air Pumps (x2)

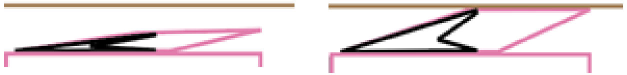
Solenoids with transistors (x4)

MPRLS sensors (x4) with I2C breakout board

Button and LED



Contact:



V ↑
P ~ constant

V constant
P ↑

Feedback Loop:

Turn on pumps

Measure pressure for each actuator

Contact?

Yes

No

All sides made contact?

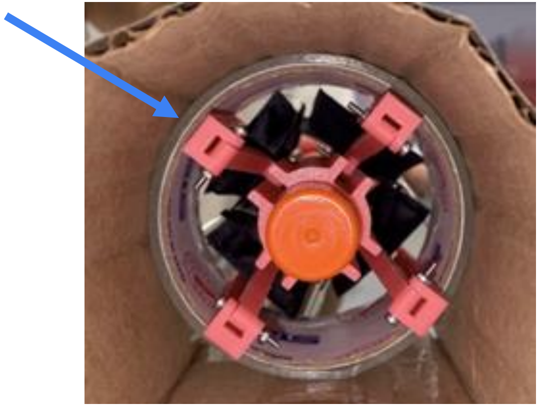
Yes

No

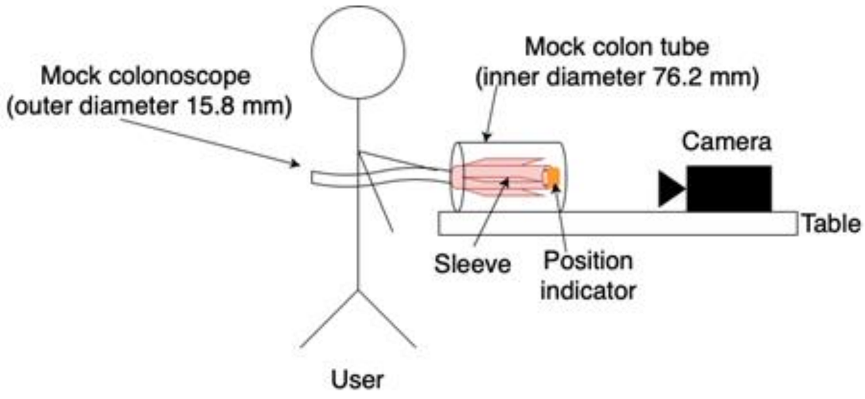
Turn off pumps

No Contact:

Contact:

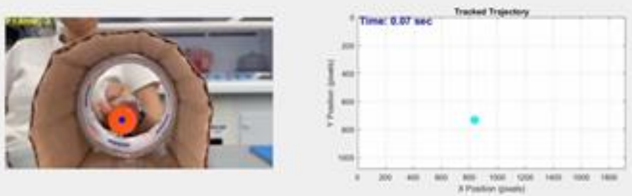


Approach - Preliminary Results

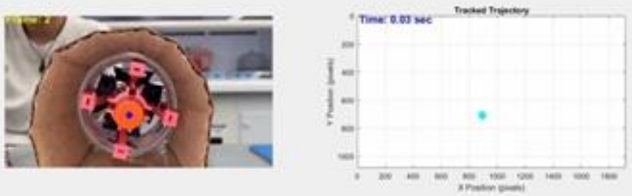


- Mock colonoscopy using flexible PVC tubing
- 5 trials of ~30 seconds each
- Manual inflation of weldable pouch motors
- Tip position detection using MATLAB

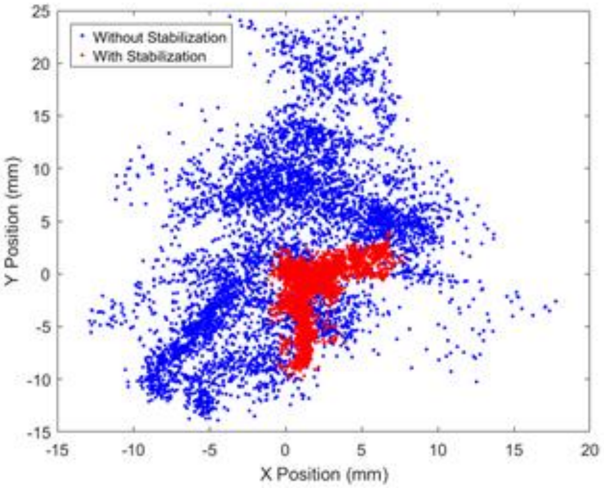
Without Stabilization:



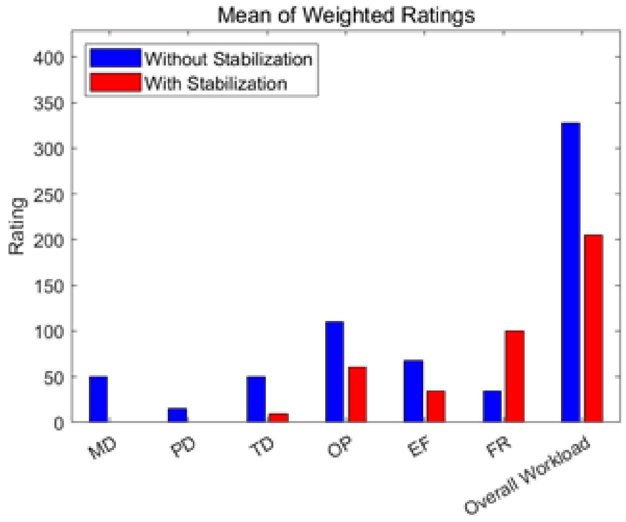
With Stabilization:



Approach - Preliminary Results



Average tip position variance without stabilization	9.09 ± 6.98 mm
Average tip position with variance stabilization	2.99 ± 2.56 mm



	Without Stabilization	With Stabilization	Percentage Change
Mental Demand	50	0	-100%
Physical Demand	15	0	-100%
Temporal Demand	50	10	-80%
Performance	110	60	-45%
Effort	67.5	35	-48%
Frustration	35	100	286%
Overall	327.5	205	**-37%**

Future Developments

Real-Time Tip Control: Joystick-based interface for tip positioning
Improved Stability and Navigation: Fine-tuned modulation of stabilization to optimize performance
Realistic Testing: Soft in-vitro or ex-vivo test setup to match wall compliance and deformation
Haptic Feedback: Potential addition of force feedback for more intuitive, surgeon-guided control
Miniaturized Design: Refine TPE bellow actuation
Off-Center Adjustment: Tip no longer constrained to centerline — adaptable to procedural needs

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